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Hydraulic fluid power — Single rod cylinders, 16 MPa (160 bar) medium and 24 MPa (250 bar) series — Mounting dimensions for accessories

Transmissions hydrauliques — Vérins 16 MPa (160 bar) série moyenne et 24 MPa (250 bar) à simple tige — Dimensions d'interchangeabilité des accessoires

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Foreword

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International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

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ISO 8132 was prepared by Technical Committee ISO/TC 131, *Fluid power systems*, Subcommittee SC 3, *Cylinders*.

This second edition cancels and replaces the first edition (ISO 8132:1986), ISO 6981:1992 and ISO 6982:1992, which have been technically revised.

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Introduction

In hydraulic fluid power systems, power is transmitted and controlled through a liquid under pressure within an enclosed circuit.

One component of such systems is the fluid power cylinder. This is a device that converts power into linear mechanical force and motion. It consists of a movable element, i.e. a piston and piston rod, operating within a cylindrical bore.

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Hydraulic fluid power — Single rod cylinders, 16 MPa (160 bar) medium and 24 MPa (250 bar) series — Mounting dimensions for accessories

1 Scope

This International Standard lays down the mounting dimensions required for the interchangeability of accessories for the 16 MPa [160 bar[1)]] medium series cylinders in accordance with ISO 6020-1 and for the 25 MPa (250 bar) series cylinders in accordance with ISO 6022. The accessories have been designed specifically for use with cylinders manufactured in accordance with ISO 6020-1 and ISO 6022, but this does not limit their application.

This International Standard covers the following accessories, identified in accordance with ISO 6099:

— AP2 — rod clevis, female thread (see Figure 1 and Table 1);

— AF3 — rod flange, circular (see Figure 2 and Table 2);

— AB4 — clevis bracket, straight (see Figure 3 and Table 3);

— AB3 — clevis bracket, in angle (see Figure 4 and Table 4);

— AT4 — trunnion bracket (see Figure 5 and Table 5);

— AA8 — pivot pin, plain or spherical bearing (cotter pin or snap ring type) (see Figure 6 and Table 6);

— AP6 — rod eye spherical, female thread (see Figure 7 and Table 7);

— AP4 — rod eye plain, female thread (see Figure 8 and Table 8).

These accessories are used on hydraulic cylinders for mechanically transmitting the cylinder force. The design of these accessories is based on the maximum forces resulting from the specified internal diameters of the cylinders and pressures according to ISO 3320 and ISO 3322.

This International Standard applies only to the dimensional criteria of products manufactured in conformity with this International Standard; it does not apply to their functional characteristics.

1) 1 bar = 0,1 MPa = 10^5 Pa; 1 MPa = 1 N/mm^2

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2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 286-2, *ISO system of limits and fits — Part 2: Tables of standard tolerance grades and limit deviations for holes and shafts*

ISO 3320, *Fluid power systems and components — Cylinder bores and piston rod diameters — Metric series*

ISO 3322, *Fluid power systems and components — Cylinders — Nominal pressures*

ISO 5598[2)], *Fluid power system and components — Vocabulary*

ISO 6020-1, *Hydraulic fluid power — Mounting dimensions for single rod cylinders, 16 MPa (160 bar) series — Part 1: Medium series*

ISO 6022[3)], *Hydraulic fluid power — Mounting dimensions for single rod cylinders, 25 MPa (250 bar) series*

ISO 6099, *Fluid power systems and components — Cylinders — Identification code for mounting dimensions and mounting types*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 5598 apply.

4 Mounting dimensions

The mounting dimensions for accessories are shown in Figures 1 to 8 and given in Tables 1 to 8.

5 Material — Load capacity

All cross-sections shall be selected so that, under the maximum tensile load produced by the cylinder, the yield strength of the material used for accessories is at least 2,5 times the maximum tensile load.

6 Application instructions

6.1 Installation

6.1.1 A tolerance of f8 in accordance with ISO 286-2 is recommended for the bearing shafts.

6.1.2 The rod clevis, female thread (AP2) shall be screwed firmly against the piston rod shoulder before the two pieces are locked.

6.2 Lubrication

6.2.1 Sufficient lubrication for the satisfactory performance of these accessories shall be provided.

2) To be published. (Revision of ISO 5598:1995)

3) To be published. (Revision of ISO 6022:1981 and ISO 8137:1986)

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6.2.2 The method and frequency of such lubrication depend on the particular operating conditions.

6.2.3 For maintenance-free mating parts, no additional lubrication is required.

7 Designation

Accessories in accordance with this International Standard shall be designated by the mounting type identification in accordance with ISO 6099, followed by "ISO 8132", followed by a dash, followed by the size type from the relevant table.

EXAMPLE 1 **AP2 ISO 8132 - 20** designates a rod clevis, female thread of type 20 ($CK = 20$) in accordance with ISO 8132.

EXAMPLE 2 **AF3 ISO 8132 - 20** designates a rod flange, circular of type 20 ($KK =$ M16 x 1,5) in accordance with ISO 8132.

EXAMPLE 3 **AB4 ISO 8132 - 20** designates a clevis bracket, straight of type 20 ($CK = 20$), in accordance with ISO 8132,

EXAMPLE 4 **AB3 ISO 8132 - 20** designates a clevis bracket, in angle of type 20 ($CK = 20$), in accordance with ISO 8132.

EXAMPLE 5 **AT4 ISO 8132 - 20** designates a trunnion bracket of type 20 ($CK = 20$), in accordance with ISO 8132.

EXAMPLE 6 **AA8 ISO 8132 - 25** designates a pivot pin, plain (cotter pin or snap ring type) of type 25 ($EK = 25$), in accordance with ISO 8132.

EXAMPLE 7 **AP6 ISO 8132 - 20** designates a rod eye spherical, female thread of type 20 ($CN = 20$), in accordance with ISO 8132.

EXAMPLE 8 **AP4 ISO 8132 - 20** designates a rod eye plain, female thread of type 20 ($CK = 20$), in accordance with ISO 8132.

8 Identification statement (Reference to this International Standard)

Use the following statement in test reports, catalogues and sales literature when electing to comply with this International Standard:

"Interchangeable cylinder accessory mounting dimensions selected in accordance with ISO 8132:2005, *Hydraulic fluid power — Single rod cylinders, 16 MPa (160 bar) medium and 25 MPa (250 bar) series — Mounting dimensions for accessories.*"

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